

AI Engineer Roadmap

Overview of the AI engineer role, responsibilities, and learning philosophy.

Key topics:

- Learn concepts deeply.
- Practice coding regularly.
- Build at least one project for each major topic.
- Revise with interview questions and documentation.

Recommended tools: Python, VS Code, Git, Jupyter, NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Docker, React.

Programming Foundations

Python, OOP, virtual environments, packaging, debugging, Git.

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Mathematics

Linear algebra, calculus, probability, statistics, optimization.

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Data Handling

NumPy, Pandas, SQL, data cleaning, visualization.

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Machine Learning

Supervised, unsupervised learning, model evaluation, feature engineering.

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Deep Learning

Neural networks, backpropagation, TensorFlow, PyTorch.

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Natural Language Processing

Tokenization, embeddings, transformers, attention.

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Computer Vision

CNNs, image preprocessing, detection, segmentation.

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Generative AI

LLMs, prompting, embeddings, RAG, fine-tuning concepts.

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LangChain

Chains, prompts, document loaders, retrievers, agents.

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LangGraph

State graphs, workflows, memory, tool calling.

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Vector Databases

FAISS, Chroma, Pinecone, indexing, retrieval.

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Model Serving

FastAPI, Docker, REST APIs, deployment.

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Cloud & MLOps

AWS/GCP/Azure, CI/CD, MLflow, monitoring.

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Frontend Basics

React, Tailwind CSS, API integration.

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Projects

Chatbots, RAG apps, recommendation systems, vision apps.

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Interview Preparation

DSA, ML theory, system design, coding.

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Learning Resources

Books, papers, documentation, courses.

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12-Month Roadmap

Month-wise progression from Python to production AI.

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Final Checklist

Portfolio, GitHub, resume, open-source, continuous learning.

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